

Functions of One Real Variable (2015–2021)

Category: Real Analysis • Official Mock Test Series

TOTAL QUESTIONS

77

TOTAL MARKS

127 M

TEST DURATION

231 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2015–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	38	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	15	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	24	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q57 • ID: JAM_2021_Q57)

NAT

+2 M

Neg: Nil (0)

Q. 57 The value of

$$\lim_{n \rightarrow \infty} \int_0^1 e^{x^2} \sin(nx) dx$$

is ____.

Question 2 (JAM 2021 • Q54 • ID: JAM_2021_Q54)

NAT

+2 M

Neg: Nil (0)

Q. 54 The largest positive number a such that

$$\int_0^5 f(x) dx + \int_0^3 f^{-1}(x) dx \geq a$$

for every strictly increasing surjective continuous function $f : [0, \infty) \rightarrow [0, \infty)$ is ____.**Question 3** (JAM 2021 • Q53 • ID: JAM_2021_Q53)

NAT

+2 M

Neg: Nil (0)

Q. 53 Consider those continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that have the property that given any $x \in \mathbb{R}$,

$$f(x) \in \mathbb{Q} \text{ if and only if } f(x+1) \in \mathbb{R} \setminus \mathbb{Q}.$$

The number of such functions is ____.

Question 4 (JAM 2021 • Q45 • ID: JAM_2021_Q45)

NAT

+1 M

Neg: Nil (0)

Q. 45 Consider the set $A = \{a \in \mathbb{R} : x^2 = a(a+1)(a+2) \text{ has a real root}\}$. The number of connected components of A is ____.

Question 5 (JAM 2021 • Q38 • ID: JAM_2021_Q38)

MSQ

+2 M

Neg: Nil (0)

JAM 2021

MATHEMATICS - MA

Q. 38 Consider the four functions from $\mathbb{R}$ to $\mathbb{R}$:

$$f_1(x) = x^4 + 3x^3 + 7x + 1, \quad f_2(x) = x^3 + 3x^2 + 4x, \quad f_3(x) = \arctan(x)$$

and

$$f_4(x) = \begin{cases} x & \text{if } x \notin \mathbb{Z}, \\ 0 & \text{if } x \in \mathbb{Z}. \end{cases}$$

Which of the following subsets of $\mathbb{R}$ are open?(A) The range of f_1 .(B) The range of f_2 .(C) The range of f_3 .(D) The range of f_4 .

Question 6 (JAM 2021 • Q35 • ID: JAM_2021_Q35)

MSQ

+2 M

Neg: Nil (0)

Q. 35 Let $f : (a, b) \rightarrow \mathbb{R}$ be a differentiable function on (a, b) . Which of the following statements is/are true?(A) $f' > 0$ in (a, b) implies that f is increasing in (a, b) .(B) f is increasing in (a, b) implies that $f' > 0$ in (a, b) .(C) If $f'(x_0) > 0$ for some $x_0 \in (a, b)$, then there exists a $\delta > 0$ such that $f(x) > f(x_0)$ for all $x \in (x_0, x_0 + \delta)$.(D) If $f'(x_0) > 0$ for some $x_0 \in (a, b)$, then f is increasing in a neighbourhood of x_0 .

Question 7 (JAM 2021 • Q32 • ID: JAM_2021_Q32)

MSQ

+2 M

Neg: Nil (0)

Q. 32 Consider the equation

$$x^{2021} + x^{2020} + \cdots + x - 1 = 0.$$

Then

(A) all real roots are positive.

(B) exactly one real root is positive.

(C) exactly one real root is negative.

(D) no real root is positive.

Question 8 (JAM 2021 • Q31 • ID: JAM_2021_Q31)

MSQ

+2 M

Neg: Nil (0)

Q. 31 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function with the property that for every $y \in \mathbb{R}$, the value of the expression

$$\sup_{x \in \mathbb{R}} [xy - f(x)]$$

is finite. Define $g(y) = \sup_{x \in \mathbb{R}} [xy - f(x)]$ for $y \in \mathbb{R}$. Then

(A) g is even if f is even.

(B) f must satisfy $\lim_{|x| \rightarrow \infty} \frac{f(x)}{|x|} = +\infty$.

(C) g is odd if f is even.

(D) f must satisfy $\lim_{|x| \rightarrow \infty} \frac{f(x)}{|x|} = -\infty$.

Question 9 (JAM 2021 • Q26 • ID: JAM_2021_Q26)

MCQ

+2 M

Neg: -0.67

Q. 26 Let $f : [0, 1] \rightarrow [0, 1]$ be a non-constant continuous function such that $f \circ f = f$. Define

$$E_f = \{x \in [0, 1] : f(x) = x\}.$$

Then

(A) E_f is neither open nor closed.

(B) E_f is an interval.

(C) E_f is empty.

(D) E_f need not be an interval.

Question 10 (JAM 2021 • Q21 • ID: JAM_2021_Q21)

MCQ

+2 M

Neg: -0.67

Q. 21 Let $f : [0, 1] \rightarrow [0, \infty)$ be a continuous function such that

$$(f(t))^2 < 1 + 2 \int_0^t f(s) ds, \text{ for all } t \in [0, 1].$$

Then

(A) $f(t) < 1 + t$ for all $t \in [0, 1]$.

(B) $f(t) > 1 + t$ for all $t \in [0, 1]$.

(C) $f(t) = 1 + t$ for all $t \in [0, 1]$.

(D) $f(t) < 1 + \frac{t}{2}$ for all $t \in [0, 1]$.

Question 11 (JAM 2021 • Q18 • ID: JAM_2021_Q18)

MCQ

+2 M

Neg: -0.67

JAM 2021

MATHEMATICS - MA

Q. 18 Consider the function

$$f(x) = \begin{cases} 1 & \text{if } x \in (\mathbb{R} \setminus \mathbb{Q}) \cup \{0\}, \\ 1 - \frac{1}{p} & \text{if } x = \frac{n}{p}, n \in \mathbb{Z} \setminus \{0\}, p \in \mathbb{N} \text{ and } \gcd(n, p) = 1. \end{cases}$$

Then

- (A) all $x \in \mathbb{Q} \setminus \{0\}$ are strict local minima for f .
- (B) f is continuous at all $x \in \mathbb{Q}$.
- (C) f is not continuous at all $x \in \mathbb{R} \setminus \mathbb{Q}$.
- (D) f is not continuous at $x = 0$.

Question 12 (JAM 2021 • Q17 • ID: JAM_2021_Q17)

MCQ

+2 M

Neg: -0.67

Q. 17 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an infinitely differentiable function such that for all $a, b \in \mathbb{R}$ with $a < b$,

$$\frac{f(b) - f(a)}{b - a} = f'\left(\frac{a + b}{2}\right).$$

Then

- (A) f must be a polynomial of degree less than or equal to 2.
- (B) f must be a polynomial of degree greater than 2.
- (C) f is not a polynomial.
- (D) f must be a linear polynomial.



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Question 13 (JAM 2021 • Q16 • ID: JAM_2021_Q16)

MCQ

+2 M

Neg: -0.67

Q. 16 Define

$$S = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \cdots \left(1 - \frac{1}{n^2}\right).$$

Then

(A) $S = 1/2$.

(B) $S = 1/4$.

(C) $S = 1$.

(D) $S = 3/4$.

Question 14 (JAM 2021 • Q7 • ID: JAM_2021_Q7)

MCQ

+1 M

Neg: -0.33

JAM 2021

MATHEMATICS - MA

Q. 7 For every $n \in \mathbb{N}$, let $f_n : \mathbb{R} \rightarrow \mathbb{R}$ be a function. From the given choices, pick the statement that is the negation of

“For every $x \in \mathbb{R}$ and for every real number $\epsilon > 0$, there exists an integer $N > 0$ such that $\sum_{i=1}^p |f_{N+i}(x)| < \epsilon$ for every integer $p > 0$.”

- (A) For every $x \in \mathbb{R}$ and for every real number $\epsilon > 0$, there does not exist any integer $N > 0$ such that $\sum_{i=1}^p |f_{N+i}(x)| < \epsilon$ for every integer $p > 0$.
- (B) For every $x \in \mathbb{R}$ and for every real number $\epsilon > 0$, there exists an integer $N > 0$ such that $\sum_{i=1}^p |f_{N+i}(x)| \geq \epsilon$ for some integer $p > 0$.
- (C) There exists $x \in \mathbb{R}$ and there exists a real number $\epsilon > 0$ such that for every integer $N > 0$, there exists an integer $p > 0$ for which the inequality $\sum_{i=1}^p |f_{N+i}(x)| \geq \epsilon$ holds.
- (D) There exists $x \in \mathbb{R}$ and there exists a real number $\epsilon > 0$ such that for every integer $N > 0$ and for every integer $p > 0$ the inequality $\sum_{i=1}^p |f_{N+i}(x)| \geq \epsilon$ holds.

Question 15 (JAM 2021 • Q4 • ID: JAM_2021_Q4)

MCQ

+1 M

Neg: -0.33

JAM 2021

MATHEMATICS - MA

Q. 4 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that for all $x \in \mathbb{R}$,

$$\int_0^1 f(xt) dt = 0. \quad (*)$$

Then

- (A) f must be identically 0 on the whole of $\mathbb{R}$.
- (B) there is an f satisfying $(*)$ that is identically 0 on $(0, 1)$ but not identically 0 on the whole of $\mathbb{R}$.
- (C) there is an f satisfying $(*)$ that takes both positive and negative values.
- (D) there is an f satisfying $(*)$ that is 0 at infinitely many points, but is not identically zero.

Question 16 (JAM 2021 • Q3 • ID: JAM_2021_Q3)

MCQ

+1 M

Neg: -0.33

Q. 3 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying $f(x) = f(x+1)$ for all $x \in \mathbb{R}$. Then

- (A) f is not necessarily bounded above.
- (B) there exists a unique $x_0 \in \mathbb{R}$ such that $f(x_0 + \pi) = f(x_0)$.
- (C) there is no $x_0 \in \mathbb{R}$ such that $f(x_0 + \pi) = f(x_0)$.
- (D) there exist infinitely many $x_0 \in \mathbb{R}$ such that $f(x_0 + \pi) = f(x_0)$.



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Question 17 (JAM 2020 • Q54 • ID: JAM_2020_Q54)

NAT

+2 M

Neg: Nil (0)

Q. 54 Let $f(x) = \sqrt{x} + \alpha x$, $x > 0$ and

$$g(x) = a_0 + a_1(x-1) + a_2(x-1)^2$$

be the sum of the first three terms of the Taylor series of $f(x)$ around $x = 1$. If $g(3) = 3$, then α is _____.

Question 18 (JAM 2020 • Q44 • ID: JAM_2020_Q44)

NAT

+1 M

Neg: Nil (0)

Q. 44 Let $S = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$ and $f : S \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{1}{x}$. Then

$$\max \left\{ \delta : \left| x - \frac{1}{3} \right| < \delta \implies \left| f(x) - f\left(\frac{1}{3}\right) \right| < 1 \right\}$$

is _____. (rounded off to two decimal places)

Question 19 (JAM 2020 • Q43 • ID: JAM_2020_Q43)

NAT

+1 M

Neg: Nil (0)

Q. 43 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that f, f', f'' are continuous functions with $f > 0, f' > 0$ and $f'' > 0$. Then

$$\lim_{x \rightarrow -\infty} \frac{f(x) + f'(x)}{2}$$

is _____.

Question 20 (JAM 2020 • Q33 • ID: JAM_2020_Q33)

MSQ

+2 M

Neg: Nil (0)

Q. 33 Let $a, b \in \mathbb{R}$ and $a < b$. Which of the following statement(s) is/are true?

- (A) There exists a continuous function $f : [a, b] \rightarrow (a, b)$ such that f is one-one
- (B) There exists a continuous function $f : [a, b] \rightarrow (a, b)$ such that f is onto
- (C) There exists a continuous function $f : (a, b) \rightarrow [a, b]$ such that f is one-one
- (D) There exists a continuous function $f : (a, b) \rightarrow [a, b]$ such that f is onto

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Question 21 (JAM 2020 • Q31 • ID: JAM_2020_Q31)

MSQ

+2 M

Neg: Nil (0)

Q. 31 Let $a, b, c \in \mathbb{R}$ such that $a < b < c$. Which of the following is/are true for any continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $f(a) = b, f(b) = c$ and $f(c) = a$?

- (A) There exists $\alpha \in (a, c)$ such that $f(\alpha) = \alpha$
- (B) There exists $\beta \in (a, b)$ such that $f(\beta) = \beta$
- (C) There exists $\gamma \in (a, b)$ such that $(f \circ f)(\gamma) = \gamma$
- (D) There exists $\delta \in (a, c)$ such that $(f \circ f \circ f)(\delta) = \delta$

Question 22 (JAM 2020 • Q28 • ID: JAM_2020_Q28)

MCQ

+2 M

Neg: -0.67

Q. 28 Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that $f\left(\frac{1}{2}\right) = -\frac{1}{2}$ and

$$|f(x) - f(y) - (x - y)| \leq \sin(|x - y|^2)$$

for all $x, y \in [0, 1]$. Then $\int_0^1 f(x) dx$ is

(A) $-\frac{1}{2}$

(B) $-\frac{1}{4}$

(C) $\frac{1}{4}$

(D) $\frac{1}{2}$

Question 23 (JAM 2020 • Q14 • ID: JAM_2020_Q14)

MCQ

+2 M

Neg: -0.67

Q. 14 Let $a \in \mathbb{R}$. If $f(x) = \begin{cases} (x+a)^2, & x \leq 0 \\ (x+a)^3, & x > 0, \end{cases}$

then

(A) $\frac{d^2 f}{dx^2}$ does not exist at $x = 0$ for any value of a

(B) $\frac{d^2 f}{dx^2}$ exists at $x = 0$ for exactly one value of a

(C) $\frac{d^2 f}{dx^2}$ exists at $x = 0$ for exactly two values of a

(D) $\frac{d^2 f}{dx^2}$ exists at $x = 0$ for infinitely many values of a

$$\int x^2 \sin \frac{1}{x} + y^2 \sin \frac{1}{y}, \quad xy \neq 0$$

Question 24 (JAM 2020 • Q3 • ID: JAM_2020_Q3)

MCQ

+1 M

Neg: -0.33

Q. 3 Which of the following is FALSE?

(A) $\lim_{x \rightarrow \infty} \frac{x}{e^x} = 0$

(B) $\lim_{x \rightarrow 0^+} \frac{1}{xe^{1/x}} = 0$

(C) $\lim_{x \rightarrow 0^+} \frac{\sin x}{1 + 2x} = 0$

(D) $\lim_{x \rightarrow 0^+} \frac{\cos x}{1 + 2x} = 0$

Question 25 (JAM 2020 • Q2 • ID: JAM_2020_Q2)

MCQ

+1 M

Neg: -0.33

Q. 2 Let $f(x) = 2x^3 - 9x^2 + 7$. Which of the following is true?

- (A) f is one-one in the interval $[-1, 1]$
 (B) f is one-one in the interval $[2, 4]$
 (C) f is NOT one-one in the interval $[-4, 0]$
 (D) f is NOT one-one in the interval $[0, 4]$

Question 26 (JAM 2019 • Q47 • ID: JAM_2019_Q47)

NAT

+1 M

Neg: Nil (0)

Q.47 If

$$g(x) = \int_{x(x-2)}^{4x-5} f(t) dt, \text{ where } f(x) = \sqrt{1+3x^4} \text{ for } x \in \mathbb{R}$$

then $g'(1) =$ _____

Question 27 (JAM 2019 • Q46 • ID: JAM_2019_Q46)

NAT

+1 M

Neg: Nil (0)

Q.46 Let $f: [0, 1] \rightarrow \mathbb{R}$ be given by

$$f(x) = \frac{\left(1+x^{\frac{1}{3}}\right)^3 + \left(1-x^{\frac{1}{3}}\right)^3}{8(1+x)}.$$

Then

$$\max \{f(x): x \in [0, 1]\} - \min \{f(x): x \in [0, 1]\}$$

is _____

Question 28 (JAM 2019 • Q45 • ID: JAM_2019_Q45)

NAT

+1 M

Neg: Nil (0)

Q.45 The coefficient of $\left(x - \frac{\pi}{2}\right)$ in the Taylor series expansion of the function

$$f(x) = \begin{cases} \frac{4(1 - \sin x)}{2x - \pi}, & x \neq \frac{\pi}{2} \\ 0, & x = \frac{\pi}{2} \end{cases}$$

about $x = \frac{\pi}{2}$, is _____

MA

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Question 29 (JAM 2019 • Q38 • ID: JAM_2019_Q38)

MSQ

+2 M

Neg: Nil (0)

Q.38 Let $f: \left(0, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ be given by

$$f(x) = (\sin x)^\pi - \pi \sin x + \pi.$$

Then which of the following statements is/are TRUE?

- (A) f is an increasing function
- (B) f is a decreasing function
- (C) $f(x) > 0$ for all $x \in \left(0, \frac{\pi}{2}\right)$
- (D) $f(x) < 0$ for some $x \in \left(0, \frac{\pi}{2}\right)$

Question 30 (JAM 2019 • Q37 • ID: JAM_2019_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.37 Let

$$f(x) = \cos(|\pi - x|) + (x - \pi) \sin |x| \text{ and } g(x) = x^2 \text{ for } x \in \mathbb{R}.$$

If $h(x) = f(g(x))$, then

- (A) h is not differentiable at $x = 0$
- (B) $h'(\sqrt{\pi}) = 0$
- (C) $h''(x) = 0$ has a solution in $(-\pi, \pi)$
- (D) there exists $x_0 \in (-\pi, \pi)$ such that $h(x_0) = x_0$

Question 31 (JAM 2019 • Q18 • ID: JAM_2019_Q18)

MCQ

+2 M

Neg: -0.67

Q.18 For which one of the following values of k , the equation

$$2x^3 + 3x^2 - 12x - k = 0$$

has three distinct real roots?

- (A) 16
- (B) 20
- (C) 26
- (D) 31

Question 32 (JAM 2019 • Q17 • ID: JAM_2019_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f'(x) > f(x)$ for all $x \in \mathbb{R}$, and $f(0) = 1$. Then $f(1)$ lies in the interval

- (A) $(0, e^{-1})$
- (B) $(e^{-1}, \sqrt{e})$
- (C) $(\sqrt{e}, e)$
- (D) (e, ∞)

Question 33 (JAM 2019 • Q16 • ID: JAM_2019_Q16)

MCQ

+2 M

Neg: -0.67

Q.16 Let $f(x) = (\ln x)^2$, $x > 0$. Then

- (A) $\lim_{x \rightarrow \infty} \frac{f(x)}{x}$ does not exist
- (B) $\lim_{x \rightarrow \infty} f'(x) = 2$
- (C) $\lim_{x \rightarrow \infty} (f(x+1) - f(x)) = 0$
- (D) $\lim_{x \rightarrow \infty} (f(x+1) - f(x))$ does not exist

Question 34 (JAM 2018 • Q55 • ID: JAM_2018_Q55)

NAT

+2 M

Neg: Nil (0)

Q.55 If $\alpha = \int_{\pi/6}^{\pi/3} \frac{\sin t + \cos t}{\sqrt{\sin 2t}} dt$, then the value of $\left(2 \sin \frac{\alpha}{2} + 1\right)^2$ is _____

Question 35 (JAM 2018 • Q52 • ID: JAM_2018_Q52)

NAT

+2 M

Neg: Nil (0)

Q.52 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be such that f'' is continuous on $\mathbb{R}$ and $f(0) = 1$, $f'(0) = 0$ and $f''(0) = -1$.

Then $\lim_{x \rightarrow \infty} \left(f \left(\sqrt{\frac{2}{x}} \right) \right)^x$ is _____ (correct up to three decimal places).

Question 36 (JAM 2018 • Q50 • ID: JAM_2018_Q50)

NAT

+1 M

Neg: Nil (0)

Q.50 The coefficient of x^4 in the power series expansion of $e^{\sin x}$ about $x = 0$ is _____ (correct up to three decimal places).

Q. 51 – Q. 60 carry two marks each

Question 37 (JAM 2018 • Q31 • ID: JAM_2018_Q31)

MSQ

+2 M

Neg: Nil (0)

Q. 31 – Q. 40 carry two marks each.

Q.31 Let $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ be defined by $f(x) = x + \frac{1}{x^3}$. On which of the following interval(s) is f one-one?

- (A) $(-\infty, -1)$ (B) $(0, 1)$ (C) $(0, 2)$ (D) $(0, \infty)$

Question 38 (JAM 2018 • Q30 • ID: JAM_2018_Q30)

MCQ

+2 M

Neg: -0.67

Q.30 For $x > \frac{-1}{2}$, let $f_1(x) = \frac{2x}{1+2x}$, $f_2(x) = \log_e(1+2x)$ and $f_3(x) = 2x$. Then which one of the following is TRUE?

- (A) $f_3(x) < f_2(x) < f_1(x)$ for $0 < x < \frac{\sqrt{3}}{2}$
- (B) $f_1(x) < f_3(x) < f_2(x)$ for $x > 0$
- (C) $f_1(x) + f_2(x) < \frac{f_3(x)}{2}$ for $x > \frac{\sqrt{3}}{2}$
- (D) $f_2(x) < f_1(x) < f_3(x)$ for $x > 0$

SECTION - B

MULTIPLE SELECT QUESTIONS (MSQ)

Q 31 – Q 40 carry two marks each

Question 39 (JAM 2018 • Q29 • ID: JAM_2018_Q29)

MCQ

+2 M

Neg: -0.67

Q.29 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function and let J be a bounded open interval in $\mathbb{R}$. Define

$$W(f, J) = \sup \{f(x) \mid x \in J\} - \inf \{f(x) \mid x \in J\}.$$

Which one of the following is FALSE?

- (A) $W(f, J_1) \leq W(f, J_2)$ if $J_1 \subset J_2$
- (B) If f is a bounded function in J and $J \supset J_1 \supset J_2 \cdots \supset J_n \supset \cdots$ such that the length of the interval J_n tends to 0 as $n \rightarrow \infty$, then $\lim_{n \rightarrow \infty} W(f, J_n) = 0$
- (C) If f is discontinuous at a point $a \in J$, then $W(f, J) \neq 0$
- (D) If f is continuous at a point $a \in J$, then for any given $\epsilon > 0$ there exists an interval $I \subset J$ such that $W(f, I) < \epsilon$

Question 40 (JAM 2018 • Q16 • ID: JAM_2018_Q16)

MCQ

+2 M

Neg: -0.67

Q.16 For $x \in \mathbb{R}$, let $f(x) = \begin{cases} x^3 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$. Then which one of the following is FALSE?

(A) $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 0$

(B) $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 0$

(C) $\frac{f(x)}{x^2}$ has infinitely many maxima and minima on the interval $(0,1)$

(D) $\frac{f(x)}{x^4}$ is continuous at $x = 0$ but not differentiable at $x = 0$

Question 41 (JAM 2018 • Q15 • ID: JAM_2018_Q15)

MCQ

+2 M

Neg: -0.67

Q.15 Suppose that $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are differentiable functions such that f is strictly increasing and g is strictly decreasing. Define $p(x) = f(g(x))$ and $q(x) = g(f(x))$, $\forall x \in \mathbb{R}$. Then, for $t > 0$, the sign of $\int_0^t p'(x) (q'(x) - 3) dx$ is

(A) positive

(B) negative

(C) dependent on t

(D) dependent on f and g

Question 42 (JAM 2018 • Q4 • ID: JAM_2018_Q4)

MCQ

+1 M

Neg: -0.33

Q.4 Let a be a positive real number. If f is a continuous and even function defined on the interval $[-a, a]$, then $\int_{-a}^a \frac{f(x)}{1+e^x} dx$ is equal to

(A) $\int_0^a f(x) dx$

(B) $2 \int_0^a \frac{f(x)}{1+e^x} dx$

(C) $2 \int_0^a f(x) dx$

(D) $2a \int_0^a \frac{f(x)}{1+e^x} dx$

Question 43 (JAM 2017 • Q57 • ID: JAM_2017_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57 For $x > 1$, let

$$f(x) = \int_1^x \left(\sqrt{\log t} - \frac{1}{2} \log \sqrt{t} \right) dt$$

The number of tangents to the curve $y = f(x)$ parallel to the line $x + y = 0$ is _____

Question 44 (JAM 2017 • Q53 • ID: JAM_2017_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53 Let $f(x) = \frac{\sin \pi x}{\pi \sin x}$, $x \in (0, \pi)$, and let $x_0 \in (0, \pi)$ be such that $f'(x_0) = 0$. Then

$$(f(x_0))^2 (1 + (\pi^2 - 1) \sin^2 x_0) = \underline{\hspace{2cm}}$$

Question 45 (JAM 2017 • Q49 • ID: JAM_2017_Q49)

NAT

+1 M

Neg: Nil (0)

Q.49 For $x > 0$, let $[x]$ denote the greatest integer less than or equal to x . Then

$$\lim_{x \rightarrow 0^+} x \left(\left\lfloor \frac{1}{x} \right\rfloor + \left\lfloor \frac{2}{x} \right\rfloor + \cdots + \left\lfloor \frac{10}{x} \right\rfloor \right) = \underline{\hspace{2cm}}$$

Question 46 (JAM 2017 • Q45 • ID: JAM_2017_Q45)

NAT

+1 M

Neg: Nil (0)

Q.45

$$\left(\int_0^1 x^4 (1-x)^5 dx \right)^{-1} = \underline{\hspace{2cm}}$$

Question 47 (JAM 2017 • Q25 • ID: JAM_2017_Q25)

MCQ

+2 M

Neg: -0.67

Q.25

$$\sum_{n=1}^{\infty} \tan^{-1} \frac{2}{n^2} =$$

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{2}$

(C) $\frac{3\pi}{4}$

(D) π

Question 48 (JAM 2017 • Q20 • ID: JAM_2017_Q20)

MCQ

+2 M

Neg: -0.67

Q.20 Let $f: \mathbb{R} \rightarrow [0, \infty)$ be a continuous function. Then which one of the following is NOT TRUE?

(A) There exists $x \in \mathbb{R}$ such that $f(x) = \frac{f(0)+f(1)}{2}$

(B) There exists $x \in \mathbb{R}$ such that $f(x) = \sqrt{f(-1)f(1)}$

(C) There exists $x \in \mathbb{R}$ such that $f(x) = \int_{-1}^1 f(t) dt$

(D) There exists $x \in \mathbb{R}$ such that $f(x) = \int_0^1 f(t) dt$

Question 49 (JAM 2017 • Q19 • ID: JAM_2017_Q19)

MCQ

+2 M

Neg: -0.67

Q.19 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(2) = 2$ and

$$|f(x) - f(y)| \leq 5(|x - y|)^{3/2}$$

for all $x \in \mathbb{R}, y \in \mathbb{R}$. Let $g(x) = x^3 f(x)$. Then $g'(2) =$

(A) 5

(B) $\frac{15}{2}$

(C) 12

(D) 24

Question 50 (JAM 2017 • Q10 • ID: JAM_2017_Q10)

MCQ

+1 M

Neg: -0.33

Q.10 If

$$f(x) = \begin{cases} 1+x & \text{if } x < 0 \\ (1-x)(px+q) & \text{if } x \geq 0 \end{cases}$$

satisfies the assumptions of Rolle's theorem in the interval $[-1, 1]$, then the ordered pair (p, q) is

- (A) $(2, -1)$ (B) $(-2, -1)$ (C) $(-2, 1)$ (D) $(2, 1)$

Question 51 (JAM 2017 • Q8 • ID: JAM_2017_Q8)

MCQ

+1 M

Neg: -0.33

Q.8 Let

$$f(x) = \frac{x + |x|(1+x)}{x} \sin\left(\frac{1}{x}\right), \quad x \neq 0.$$

Write $L = \lim_{x \rightarrow 0^-} f(x)$ and $R = \lim_{x \rightarrow 0^+} f(x)$. Then which one of the following is TRUE?

- (A) L exists but R does not exist
 (B) L does not exist but R exists
 (C) Both L and R exist
 (D) Neither L nor R exists

Question 52 (JAM 2017 • Q7 • ID: JAM_2017_Q7)

MCQ

+1 M

Neg: -0.33

Q.7

Let $f_1(x), f_2(x), g_1(x), g_2(x)$ be differentiable functions on $\mathbb{R}$. Let $F(x) = \begin{vmatrix} f_1(x) & f_2(x) \\ g_1(x) & g_2(x) \end{vmatrix}$ be the determinant of the matrix $\begin{bmatrix} f_1(x) & f_2(x) \\ g_1(x) & g_2(x) \end{bmatrix}$. Then $F'(x)$ is equal to

- (A) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2'(x) & g_2(x) \end{vmatrix}$
 (B) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2(x) & g_2'(x) \end{vmatrix}$
 (C) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} - \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2(x) & g_2'(x) \end{vmatrix}$
 (D) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1'(x) & g_2'(x) \end{vmatrix}$

Question 53 (JAM 2017 • Q2 • ID: JAM_2017_Q2)

MCQ

+1 M

Neg: -0.33

Q.2 Let $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that φ' is strictly increasing with $\varphi'(1) = 0$. Let α and β denote the minimum and maximum values of $\varphi(x)$ on the interval $[2, 3]$, respectively. Then which one of the following is TRUE?

- (A) $\beta = \varphi(3)$ (B) $\alpha = \varphi(2.5)$ (C) $\beta = \varphi(2.5)$ (D) $\alpha = \varphi(3)$

Question 54 (JAM 2016 • Q49 • ID: JAM_2016_Q49)

NAT

+1 M

Neg: Nil (0)

Q.49 Let $f: (0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that

$$\int_0^x f(t) dt = -2 + \frac{x^2}{2} + 4x \sin 2x + 2 \cos 2x.$$

Then the value of $\frac{1}{\pi} f\left(\frac{\pi}{4}\right)$ is _____

Question 55 (JAM 2016 • Q46 • ID: JAM_2016_Q46)

NAT

+1 M

Neg: Nil (0)

Q.46 If $f: (-1, \infty) \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{1+x}$ is expressed as

$$f(x) = \frac{2}{3} + \frac{1}{9}(x-2) + \frac{c(x-2)^2}{(1+\xi)^3},$$

where ξ lies between 2 and x , then the value of c is _____

Question 56 (JAM 2016 • Q37 • ID: JAM_2016_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.37 Let $P(x) = \left(\frac{5}{13}\right)^x + \left(\frac{12}{13}\right)^x - 1$ for all $x \in \mathbb{R}$. Then which of the following statement(s) is(are) TRUE?

- (A) The equation $P(x) = 0$ has exactly one solution in $\mathbb{R}$
- (B) $P(x)$ is strictly increasing for all $x \in \mathbb{R}$
- (C) The equation $P(x) = 0$ has exactly two solutions in $\mathbb{R}$
- (D) $P(x)$ is strictly decreasing for all $x \in \mathbb{R}$

MA

9/13

Question 57 (JAM 2016 • Q32 • ID: JAM_2016_Q32)

MSQ

+2 M

Neg: Nil (0)

Q.32 The value(s) of the integral

$$\int_{-\pi}^{\pi} |x| \cos nx \, dx, \quad n \geq 1$$

is (are)

- (A) 0 when n is even
- (B) 0 when n is odd
- (C) $-\frac{4}{n^2}$ when n is even
- (D) $-\frac{4}{n^2}$ when n is odd

MA

8/13

Question 58 (JAM 2016 • Q30 • ID: JAM_2016_Q30)

MCQ

+2 M

Neg: -0.67

Q.30 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} x(1 + x^\alpha \sin(\ln x^2)) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Then, at $x = 0$, the function f is

- (A) continuous and differentiable when $\alpha = 0$
- (B) continuous and differentiable when $\alpha > 0$
- (C) continuous and differentiable when $-1 < \alpha < 0$
- (D) continuous and differentiable when $\alpha < -1$

SECTION - B

MULTIPLE SELECT QUESTIONS (MSQ)

Q. 31 – Q. 40 carry two marks each.

Question 59 (JAM 2016 • Q25 • ID: JAM_2016_Q25)

MCQ

+2 M

Neg: -0.67

Q.25 Let $f(x) = \frac{1}{1+|x|} + \frac{1}{1+|x-1|}$ for all $x \in [-1, 1]$. Then which one of the following is TRUE?

- (A) Maximum value of $f(x)$ is $\frac{3}{2}$
- (B) Minimum value of $f(x)$ is $\frac{1}{3}$
- (C) Maximum of $f(x)$ occurs at $x = \frac{1}{2}$
- (D) Minimum of $f(x)$ occurs at $x = 1$

Question 60 (JAM 2016 • Q14 • ID: JAM_2016_Q14)

MCQ

+2 M

Neg: -0.67

Q.14 The value of the integral

$$\frac{(2n)!}{2^{2n} (n!)^2} \int_{-1}^1 (1-x^2)^n dx, \quad n \in \mathbb{N}$$

is

(A) $\frac{2}{(2n+1)!}$

(B) $\frac{2n}{(2n+1)!}$

(C) $\frac{2 (n!)}{2n+1}$

(D) $\frac{(n+1)!}{2n+1}$

Question 61 (JAM 2016 • Q6 • ID: JAM_2016_Q6)

MCQ

+1 M

Neg: -0.33

Q.6 The value of $\lim_{(x,y) \rightarrow (2,-2)} \frac{\sqrt{(x-y)}-2}{x-y-4}$ is

(A) 0

(B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

Question 62 (JAM 2016 • Q5 • ID: JAM_2016_Q5)

MCQ

+1 M

Neg: -0.33

Q.5 Let f be a strictly monotonic continuous real valued function defined on $[a, b]$ such that $f(a) < a$ and $f(b) > b$. Then which one of the following is TRUE?

- (A) There exists exactly one $c \in (a, b)$ such that $f(c) = c$
 (B) There exist exactly two points $c_1, c_2 \in (a, b)$ such that $f(c_i) = c_i$, $i = 1, 2$
 (C) There exists no $c \in (a, b)$ such that $f(c) = c$
 (D) There exist infinitely many points $c \in (a, b)$ such that $f(c) = c$

MA

3/13

Question 63 (JAM 2016 • Q3 • ID: JAM_2016_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 Let $f: [-1, 1] \rightarrow \mathbb{R}$ be a continuous function. Then the integral

$$\int_0^{\pi} x f(\sin x) dx$$

is equivalent to

(A)

$$\frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$$

(B)

$$\frac{\pi}{2} \int_0^{\pi} f(\cos x) dx$$

(C)

$$\pi \int_0^{\pi} f(\cos x) dx$$

(D)

$$\pi \int_0^{\pi} f(\sin x) dx$$

Question 64 (JAM 2015 • Q59 • ID: JAM_2015_Q59)

NAT

+2 M

Neg: Nil (0)

Q.19 The limit

$$\lim_{x \rightarrow 0^+} \frac{9}{x} \left(\frac{1}{\tan^{-1} x} - \frac{1}{x} \right)$$

is equal to _____

Question 65 (JAM 2015 • Q56 • ID: JAM_2015_Q56)

NAT

+2 M

Neg: Nil (0)

Q.16

The coefficient of $\left(x - \frac{\pi}{4}\right)^3$ in the Taylor series expansion of the function

$$f(x) = 3 \sin x \cos\left(x + \frac{\pi}{4}\right), \quad x \in \mathbb{R}$$

about the point $\frac{\pi}{4}$, correct up to three decimal places, is _____

Question 66 (JAM 2015 • Q53 • ID: JAM_2015_Q53)

NAT

+2 M

Neg: Nil (0)

Q.13 The limit

$$\lim_{n \rightarrow \infty} \sum_{k=2}^n \frac{1}{k^3 - k}$$

is equal to _____

Question 67 (JAM 2015 • Q50 • ID: JAM_2015_Q50)

NAT

+1 M

Neg: Nil (0)

Q.10

Let $f: (0, 1) \rightarrow \mathbb{R}$ be a continuously differentiable function such that f' has finitely many zeros in $(0, 1)$ and f' changes sign at exactly two of these points. Then for any $y \in \mathbb{R}$, the maximum number of solutions to $f(x) = y$ in $(0, 1)$ is _____

Q. 11 – Q. 20 carry two marks each

Question 68 (JAM 2015 • Q49 • ID: JAM_2015_Q49)

NAT

+1 M

Neg: Nil (0)

Q.9

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x^6 - 1, & x \in \mathbb{Q} \\ 1 - x^6, & x \notin \mathbb{Q} \end{cases}$$

The number of points at which f is continuous, is _____

Question 69 (JAM 2015 • Q40 • ID: JAM_2015_Q40)

MSQ

+2 M

Neg: Nil (0)

Q.10

Let $f: (-1, 1) \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = x^2 e^{1/(1-x^2)}$$

Then

- (A) f is decreasing in $(-1, 0)$ (B) f is increasing in $(0, 1)$
- (C) $f(x) = 1$ has two solutions in $(-1, 1)$ (D) $f(x) = 1$ has no solutions in $(-1, 1)$

MA

13/16

Question 70 (JAM 2015 • Q39 • ID: JAM_2015_Q39)

MSQ

+2 M

Neg: Nil (0)

Q.9

Let $f, g: [0, 1] \rightarrow [0, 1]$ be functions. Let $R(f)$ and $R(g)$ be the ranges of f and g , respectively. Which of the following statements is (are) true?

- (A) If $f(x) \leq g(x)$ for all $x \in [0, 1]$, then $\sup R(f) \leq \inf R(g)$
- (B) If $f(x) \leq g(x)$ for some $x \in [0, 1]$, then $\inf R(f) \leq \sup R(g)$
- (C) If $f(x) \leq g(y)$ for some $x, y \in [0, 1]$, then $\inf R(f) \leq \sup R(g)$
- (D) If $f(x) \leq g(y)$ for all $x, y \in [0, 1]$, then $\sup R(f) \leq \inf R(g)$

Question 71 (JAM 2015 • Q37 • ID: JAM_2015_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.7 Which of the following statements is (are) true on the interval $(0, \frac{\pi}{2})$?

- (A) $\cos x < \cos(\sin x)$ (B) $\tan x < x$
- (C) $\sqrt{1+x} < 1 + \frac{x}{2} - \frac{x^2}{8}$ (D) $\frac{1-x^2}{2} < \ln(2+x)$

Question 72 (JAM 2015 • Q31 • ID: JAM_2015_Q31)

MSQ

+2 M

Neg: Nil (0)

Q.1

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \int_{-5}^x (t-1)^3 dt$.
In which of the following interval(s), f takes the value 1?

- (A) $[-6, 0]$ (B) $[-2, 4]$ (C) $[2, 8]$ (D) $[6, 12]$

Question 73 (JAM 2015 • Q28 • ID: JAM_2015_Q28)

MCQ

+2 M

Neg: -0.67

Q.28

For $m, n \in \mathbb{N}$, define $f_{m,n}(x) = \begin{cases} x^m \sin\left(\frac{1}{x^n}\right), & x \neq 0 \\ 0 & x = 0 \end{cases}$

Then at $x = 0$, $f_{m,n}$ is

- (A) differentiable for each pair m, n with $m > n$
(B) differentiable for each pair m, n with $m < n$
(C) not differentiable for each pair m, n with $m > n$
(D) not differentiable for each pair m, n with $m < n$

Question 74 (JAM 2015 • Q27 • ID: JAM_2015_Q27)

MCQ

+2 M

Neg: -0.67

Q.27

For $n \geq 2$, let $f_n: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f_n(x) = x^n \sin x$. Then at $x = 0$, f_n has a

- (A) local maximum if n is even
(B) local maximum if n is odd
(C) local minimum if n is even
(D) local minimum if n is odd

Question 75 (JAM 2015 • Q21 • ID: JAM_2015_Q21)

MCQ

+2 M

Neg: -0.67

Q.21 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a strictly increasing continuous function. If $\{a_n\}$ is a sequence in $[0, 1]$, then the sequence $\{f(a_n)\}$ is

- (A) increasing (B) bounded
(C) convergent (D) not necessarily bounded

Question 76 (JAM 2015 • Q17 • ID: JAM_2015_Q17)

MCQ

+2 M

Neg: -0.67

Q.17 The limit

$$\lim_{x \rightarrow 0^+} \frac{1}{\sin^2 x} \int_{\frac{x}{2}}^x \sin^{-1} t \, dt$$

is equal to

- (A) 0 (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{3}{8}$

Question 77 (JAM 2015 • Q7 • ID: JAM_2015_Q7)

MCQ

+1 M

Neg: -0.33

Q.7 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 0$. If for all $x \in \mathbb{R}$, $1 < f'(x) < 2$, then which one of the following statements is true on $(0, \infty)$?

- (A) f is unbounded (B) f is increasing and bounded
(C) f has at least one zero (D) f is periodic

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Functions of One Real Variable (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	0
2	NAT	+2	15
3	NAT	+2	0
4	NAT	+1	2
5	MSQ	+2	B, C, D
6	MSQ	+2	A, C
7	MSQ	+2	A, B
8	MSQ	+2	A, B
9	MCQ	+2	B
10	MCQ	+2	A
11	MCQ	+2	A
12	MCQ	+2	A
13	MCQ	+2	A
14	MCQ	+1	C
15	MCQ	+1	A
16	MCQ	+1	D
17	NAT	+2	0.5 to 0.5
18	NAT	+1	0.08 to 0.09
19	NAT	+1	0 to INF
20	MSQ	+2	A;C;D
21	MSQ	+2	A;C;D
22	MCQ	+2	A
23	MCQ	+2	A
24	MCQ	+1	D
25	MCQ	+1	D
26	NAT	+1	8 to 8

Q. No.	Type	Marks	Official Key
27	NAT	+1	0.25 to 0.25
28	NAT	+1	1 to 1
29	MSQ	+2	B, C
30	MSQ	+2	B, C, D
31	MCQ	+2	A
32	MCQ	+2	D
33	MCQ	+2	C
34	NAT	+2	2.9 to 3.1
35	NAT	+2	0.350 to 0.380
36	NAT	+1	−0.130 to −0.120
37	MSQ	+2	B
38	MCQ	+2	Marks All to
39	MCQ	+2	B
40	MCQ	+2	D
41	MCQ	+2	A
42	MCQ	+1	A
43	NAT	+2	0.9 to 1.1
44	NAT	+2	0.9 to 1.1
45	NAT	+1	54.9 to 55.1
46	NAT	+1	1259.9 to 1260.1
47	MCQ	+2	C
48	MCQ	+2	C
49	MCQ	+2	D
50	MCQ	+1	D
51	MCQ	+1	A
52	MCQ	+1	B

Q. No.	Type	Marks	Official Key
53	MCQ	+1	A
54	NAT	+1	0.25 to 0.25
55	NAT	+1	−1 to −1
56	MSQ	+2	A;D
57	MSQ	+2	A;D
58	MCQ	+2	MTA
59	MCQ	+2	A
60	MCQ	+2	C
61	MCQ	+1	B
62	MCQ	+1	A
63	MCQ	+1	A
64	NAT	+2	3
65	NAT	+2	1.41 to 1.42
66	NAT	+2	0.25
67	NAT	+1	3
68	NAT	+1	2
69	MSQ	+2	A;B;C
70	MSQ	+2	B;C;D
71	MSQ	+2	A;D
72	MSQ	+2	A;C;D
73	MCQ	+2	A
74	MCQ	+2	D
75	MCQ	+2	B
76	MCQ	+2	D
77	MCQ	+1	A

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.

- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.